Monsoon Rainfall & Flood-Risk Analysis for Multan

Using ERA5 Reanalysis Data

A 10-Year Trend Analysis (2016-2025) with a Focused Case Study of the 2022 Flood Year

Project Report

Data source: Copernicus Climate Data Store (CDS) - ERA5 hourly reanalysis, single levels

Region: Multan, Punjab, Pakistan (30.0-30.5N, 71.2-71.8E)

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1. Project Overview

This report documents a complete analysis of monsoon-season rainfall and temperature behavior in Multan, Pakistan, built from ERA5 reanalysis data. The project has two linked goals: (1) characterize how monsoon rainfall (June-September) behaved across ten recent years, 2016-2025, including whether any statistically meaningful trend exists; and (2) examine 2022 - Pakistan's worst recent flood year - as a detailed case study, comparing it against the ten-year norm across several specific indices of rainfall intensity and frequency.

Multan is a flood-vulnerable district in a monsoon-dependent region, and the 2022 floods caused severe, well-documented damage across Pakistan. The purpose of this analysis is to ground that event in real, defensible statistics - not just to confirm that 2022 was unusual, but to characterize precisely how it was unusual, and whether the broader ten-year record shows any warning signal of worsening conditions.

Scope of this analysis

This is a rainfall and temperature analysis using reanalysis data averaged over a small grid box around Multan. It does not include flood routing, terrain/elevation effects, drainage modeling, or climate-model bias correction - all deliberately out of scope. See Section 6 (Limitations) for the full list of what this analysis does and does not claim.

2. Data Source & Methodology

2.1 What is ERA5, and what is a "reanalysis"?

ERA5 is a global atmospheric reanalysis dataset produced by the European Centre for Medium-Range Weather Forecasts (ECMWF). A "reanalysis" is not a live weather observation and not a forecast - it is a physically consistent, best-estimate reconstruction of the atmosphere's past state, built by combining a weather model with millions of historical real-world observations (satellites, weather stations, balloons, ships, etc.) through a process called data assimilation. The result is a complete, gap-free hourly record stretching back decades, available at roughly 0.25-degree (about 25-31km) grid resolution worldwide.

ERA5 data is distributed through the Copernicus Climate Data Store (CDS), a free public service operated by the European Union's Copernicus Climate Change Service. All data used in this project was retrieved from CDS via its API.

2.2 Variables, Region, and Time Range

Variable	CDS name	Unit (raw / converted)	Type
Total precipitation	total_precipitation	meters / millimeters	Accumulated (see Sec. 3.2)
2-meter temperature	2m_temperature	Kelvin / Celsius	Instantaneous

Region: a bounding box of 30.5N, 71.2E to 30.0N, 71.8E (North, West, South, East), centered tightly on Multan city (approximately 30.16N, 71.52E). At ERA5's native 0.25-degree grid, this produces a small 3x3 grid of points immediately surrounding the city, which are averaged together to produce a single representative time series for the region (see Section 3.3).

Time range: hourly data for the monsoon months (June-September) across 2016-2021 and 2023-2025 (9 years), plus every month of 2022 in full, to support the flood case study's need for a full-year comparison rather than a monsoon-only slice.

2.3 Tools & Software

Tool	Purpose
Google Colab	Cloud notebook environment - no local install needed
Google Drive	Persistent file storage (Colab's own local disk is wiped on disconnect)
cdsapi	Official Python client for requesting data from the Copernicus Climate Data Store
xarray	Handles multi-dimensional gridded climate data (time x latitude x longitude)
pandas	Tabular data handling - daily/yearly aggregation, summary tables
numpy	Numerical operations supporting the above
scipy.stats (linregress)	Linear regression and statistical significance testing for the trend analysis
matplotlib	All charts and maps in this report

2.4 Data Storage & Reproducibility

All downloaded files and processed outputs are saved to a Google Drive folder (ERA5_Multan) rather than Colab's local disk, since Colab erases local files whenever the runtime disconnects - which happens automatically, even from brief inactivity. Every download step checks whether its target file already exists before requesting it again, so an interrupted run can simply be re-run without wasting time or CDS quota re-downloading completed years.

3. Technical Process

3.1 Downloading the Data

Data was requested year by year through the cdsapi client, in netCDF format. Each request specifies the variable list, year, months, all days, all 24 hours, and the bounding box described above. Requesting all ten years in one call was avoided in favor of a per-year loop, both to make partial failures easy to resume and to keep the resulting files manageably sized.

3.2 A Key Technical Challenge: Accumulated vs. Instantaneous Variables

One genuine complication surfaced during this project and is worth explaining, since it affects how the rainfall numbers in this report were derived. ERA5's variables fall into two different categories:

- Instantaneous variables (e.g. 2m temperature) represent the value at a single moment in time - a snapshot.
- Accumulated variables (e.g. total precipitation) represent a running total building up since a reference time, not a value at an instant.

When both variable types are requested together and converted to netCDF, the Copernicus conversion service splits them into two separate internal files (one for each type), then bundles both inside a single zip archive automatically. Left unhandled, this can cause a data-loading tool to silently keep only one variable and drop the other. This project's download code explicitly unpacks that zip, merges the two internal files back into one clean dataset per year, and verifies afterward that both variables are present with zero missing values (confirmed in Section 3.3) - so this complication does not affect the results, but it is a genuine ERA5/CDS quirk worth documenting for anyone continuing this work.

3.3 From Raw Grid to Daily Numbers

Once loaded, the raw hourly grid data was processed through a consistent pipeline:

- Spatial averaging: the 3x3 grid of points around Multan was averaged into a single hourly value per variable, producing one representative time series for the region rather than nine separate ones.
- Unit conversion: precipitation was converted from meters to millimeters (x1000); temperature was converted from Kelvin to Celsius (-273.15).
- Time zone conversion: all timestamps were shifted from UTC to Pakistan Standard Time (UTC+5), since a diurnal ("time of day") analysis is only meaningful in local time.
- Daily aggregation: hourly values were rolled up into daily totals - rainfall is summed (a day's total rainfall is the sum of each hour's rainfall), while temperature is averaged (a day's average temperature is the mean of each hour's reading), since these are the physically correct aggregation methods for each variable type.

A data-quality check confirmed the final dataset covers 2016-06-01 through 2025-09-30 with zero missing values across 316,008 individual hourly grid-cell readings for both rainfall and temperature - the full record is complete.

3.4 Statistical Methods

Heavy rain day: any day with more than 10mm of total rainfall - a standard threshold used in meteorology to flag meaningfully intense rainfall (as opposed to trace or light rain).

Dry spell: the longest run of consecutive days with less than 1mm of rainfall within a monsoon season - a standard proxy for drought stress or a prolonged break in monsoon activity.

Linear trend (slope, R-squared, p-value): a straight line was fit through the ten yearly rainfall totals using linear regression. Three numbers describe how much that line should be trusted: the slope tells us the average rate of change (mm per year) the fitted line implies; R-squared (between 0 and 1) tells us what fraction of the year-to-year variation the line actually explains, with values near 0 meaning the line explains almost nothing and the data looks more like noise; and the p-value tells us the probability of seeing a slope this steep purely by chance if there were truly no trend at all - by scientific convention, a p-value above 0.05 means the result cannot be distinguished from random noise, i.e. no statistically defensible trend exists.

95th-percentile extreme-day threshold: an alternative, purely data-driven way of defining an "extreme" rainfall day - rather than picking a fixed number like 10mm, the threshold is set at whatever value only the top 5% of all days in the entire ten-year record exceed. This complements the fixed 10mm heavy-rain-day threshold with a self-calibrating measure of extremity.

4. Results & Interpretation

4.1 Ten-Year Rainfall Totals

Year	Monsoon Total (mm)
2016	186.8
2017	218.2
2018	84.7
2019	224.6
2020	252.3
2021	53.4
2022	374.2
2023	145.1
2024	274.7
2025	190.8

Monsoon rainfall totals swing widely from year to year, ranging from a low of about 53mm (2021) to a high of about 374mm (2022) - roughly a seven-fold difference between the driest and wettest years in the record. There is no obvious steady upward or downward march across the decade by eye; Section 4.2 tests this formally.

4.2 Is Rainfall Increasing? The Trend Analysis

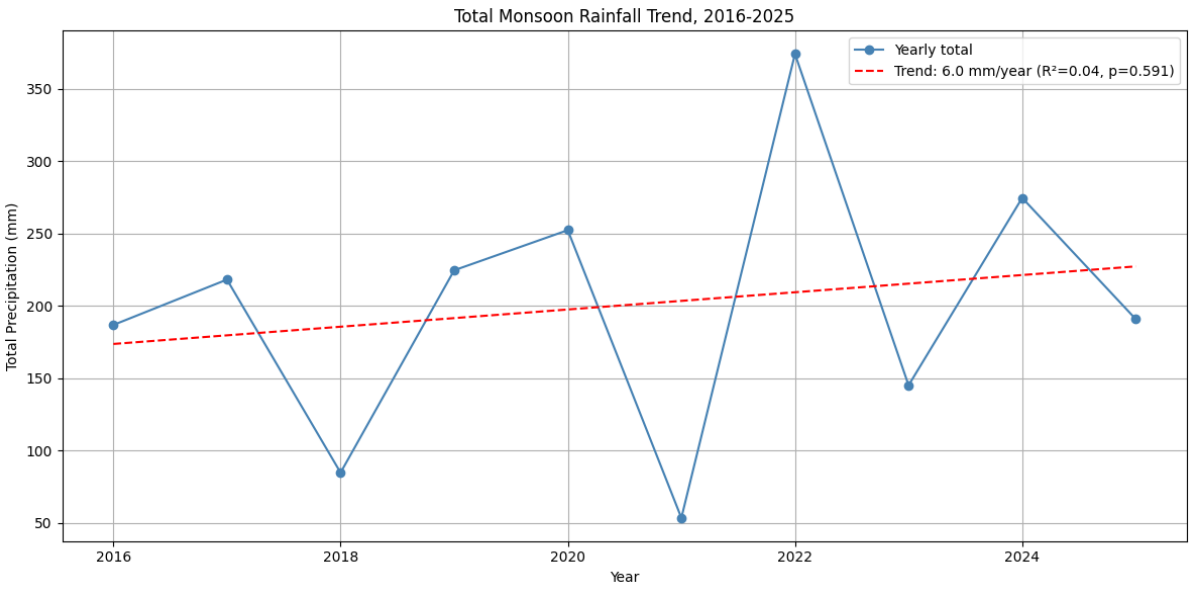


Figure 1. Total monsoon rainfall by year with fitted linear trend line.

Result: no statistically significant trend

Slope = +5.96 mm/year | R-squared = 0.038 | p-value = 0.591. The fitted line suggests rainfall is increasing by about 6mm per year, but this should not be reported as a real trend. An R-squared of 0.038 means year-to-year variation explains under 4% of what the line is fitting - essentially none. A p-value of 0.591 is far above the conventional 0.05 significance threshold, meaning a slope this steep would be unsurprising to see by pure chance even if the true underlying trend were exactly zero. This matches this project's own stated limitation: a 10-year record is too short for formal climate trend detection (30+ years is the typical scientific standard). The honest, defensible finding is: no significant trend - what the record shows instead is high year-to-year volatility, visible directly in the alternating peaks and troughs in Figure 1.

4.3 Extreme Rainfall Indices per Year

Year	Wettest Day	Wettest Day (mm)	Heavy Rain Days (>10mm)	Longest Dry Spell (days)
2016	27 Aug	22.2	5	16
2017	28 Jun	26.5	8	18
2018	20 Jul	7.9	0	23
2019	10 Aug	19.3	7	26
2020	31 Aug	104.3	5	26
2021	13 Jul	7.2	0	30
2022	21 Jun	46.7	10	36
2023	07 Jul	22.4	5	50
2024	17 Aug	42.7	9	25
2025	08 Sep	29.5	7	25

Two results stand out here beyond the 2022 case study itself. First, 2020's wettest single day - 104.3mm on 31 August - is the single most intense rainfall day anywhere in the entire ten-year record, more than double 2022's peak day, yet 2020 was not a flood year. Second, the longest dry spells show a rough upward drift across the decade (16 days in 2016 up to 50 days in 2023) before dropping back down in 2024-2025; with only ten noisy data points, this pattern has not been tested for statistical significance the way total rainfall was in Section 4.2, and should be read as a descriptive observation rather than a confirmed trend.

4.4 2022 Flood Case Study: How Different Was It?

Metric	2022	Average of other 9 years
Total monsoon rainfall	374.2 mm	181.2 mm
Wettest single day	46.7 mm	31.3 mm
Heavy rain days (>10mm)	10	5.1
Longest dry spell	36 days	26.6 days

Interpretation

2022 stands out clearly on total volume (more than double the other years' average) and on the frequency of heavy-rain days (nearly double), but its single wettest day was not the most extreme in the whole dataset - 2020 holds that record. This is a meaningful distinction for flood risk: 2022 looks less like the result of one catastrophic downpour and more like the product of persistent, repeated heavy rain across the season. That pattern matters physically, because each successive storm falls on already-saturated ground, compounding runoff and flood risk in a way a single isolated extreme event would not. The elevated dry-spell figure (36 days, the longest of the decade alongside 2023) further suggests 2022's season may have been front- or back-loaded around its rain clusters, rather than delivering rainfall evenly.

4.5 Spatial Patterns Across the Multan Region

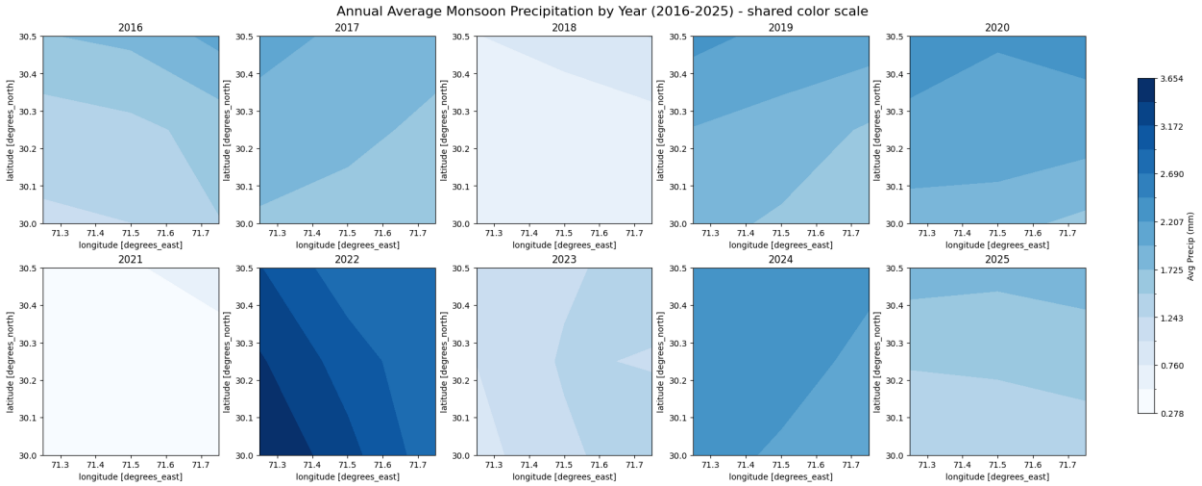


Figure 2. Annual average monsoon precipitation, all ten years shown on one shared color scale.

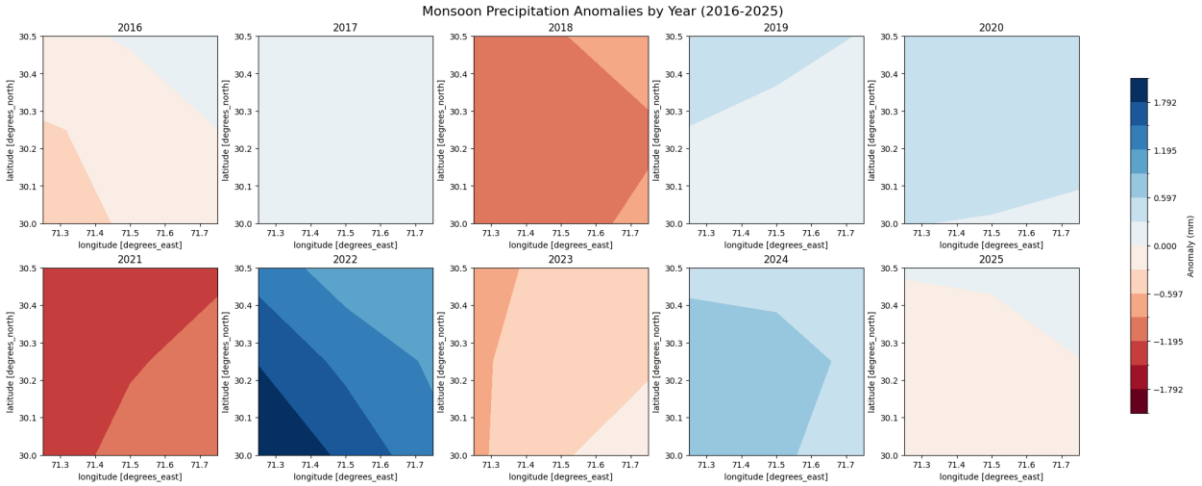


Figure 3. Monsoon precipitation anomalies (deviation from the ten-year average), shared color scale.

Figure 2 uses one shared color scale across every year (an earlier draft of this chart let each year auto-scale independently, which made colors visually misleading between panels - that has been corrected here). With a consistent scale, 2022's panel is unambiguously the darkest (wettest) and 2021's is the lightest (driest) of the decade, matching the numbers in Section 4.1.

Figure 3, the anomaly view, makes the decade's real story easier to see: 2021 (deep red, well below normal) is just as extreme in its own direction as 2022 (deep blue, well above normal). Most other years sit much closer to the ten-year average. This decade did not drift toward more rain overall - it swung between one severe drought year and one severe flood year, which reinforces the Section 4.2 finding that volatility, not a directional trend, is the defining feature of this record.

A secondary spatial pattern is visible within most individual years too: rainfall tends to be slightly higher toward the southwest corner of the study box than the northeast corner (most visible in 2016, 2019, 2020, and 2022). Given the box only spans about 55km corner to corner, this is a subtle, local gradient rather than a distinct regional climate difference, but it is consistent enough across years to be worth noting.

4.6 When Does It Rain? Diurnal Pattern

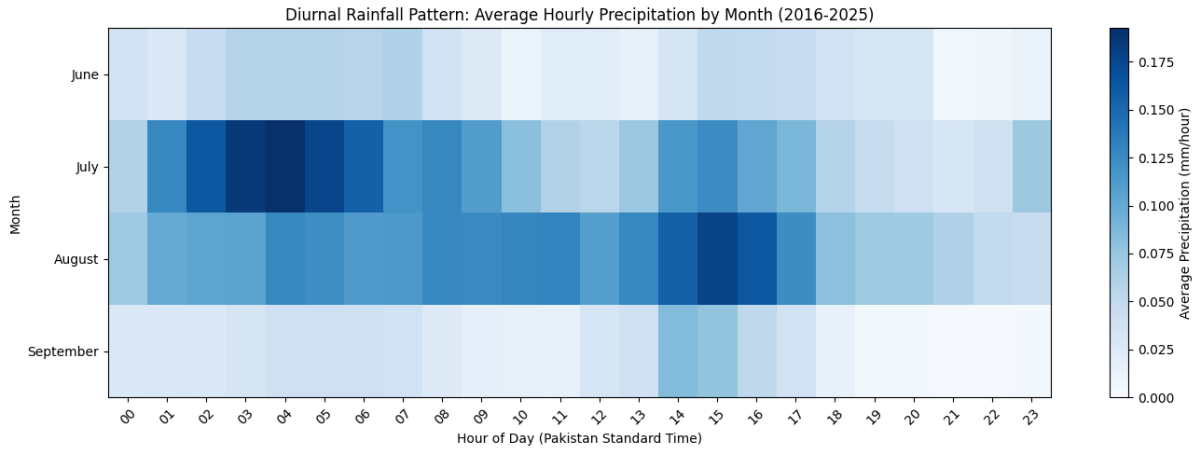


Figure 4. Average hourly rainfall by month and hour of day, Pakistan Standard Time.

Across all monsoon months combined, the single hour with the highest average rainfall is 15:00 PKT (mid-afternoon) - consistent with the classic pattern of late-afternoon convective monsoon storms common across South Asia, where daytime heating builds the instability that triggers rain.

Looking month by month reveals a more specific and genuinely interesting pattern than the overall average alone suggests: July's peak rainfall hour is 04:00 (pre-dawn), while August's peak is 15:00 (mid-afternoon) - visible in Figure 4 as two distinct dark bands sitting in different parts of the day. June peaks at 07:00 and September at 14:00. This suggests the two wettest months of the season may be driven by somewhat different rainfall mechanisms - July's early-morning peak points toward larger-scale overnight monsoon systems, while August's afternoon peak points toward local daytime convective storms - though confirming that distinction would need meteorological analysis beyond this project's scope.

4.7 Extreme Event Days (95th-Percentile Method)

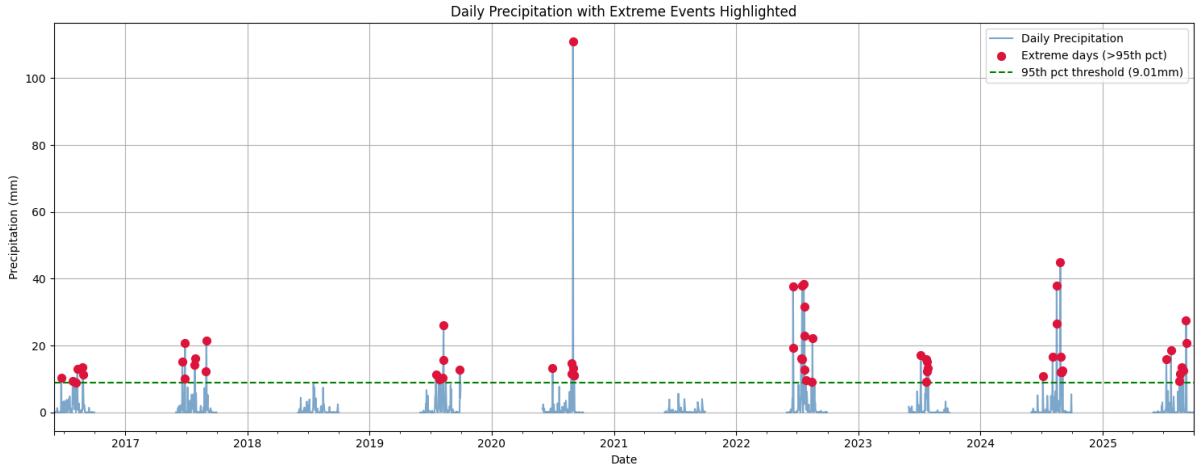


Figure 5. Daily rainfall with statistically extreme days (top 5% of all days) highlighted.

The self-calibrating 95th-percentile threshold works out to 9.01mm/day - close to, but distinct from, the fixed 10mm "heavy rain day" threshold used elsewhere in this report. 61 days across the ten-year record exceed it. The five most extreme days in the entire dataset were: 31 Aug 2020 (110.9mm), 27 Aug 2024 (44.9mm), 22 Jul 2022 (38.4mm), 16 Aug 2024 (38.0mm), and 17 Jul 2022 (37.8mm). Notably, 2018 recorded zero days above this threshold - the only year in the record to do so - reinforcing 2018's position as one of the two clearly drought-leaning years alongside 2021.

4.8 Temperature: Was 2022 Hotter? (A Corrected Interpretation)

Correction from an earlier draft

An earlier draft of this analysis assumed, without checking, that 2022's flood period would coincide with above-average heat, based on the general mechanism that intense pre-monsoon heat can strengthen the monsoon low-pressure system. The actual data does not support that claim for the monsoon-season average, and this section replaces that earlier assumption with what the numbers actually show.

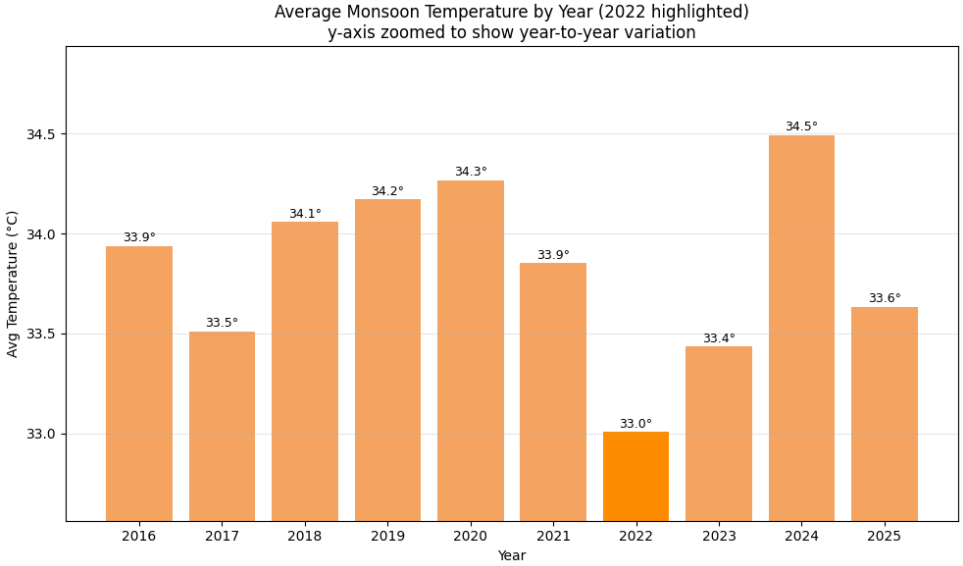


Figure 6. Average monsoon-season temperature by year (2022 highlighted); y-axis zoomed to show year-to-year variation.

2022's average monsoon temperature was 33.0C - the coolest of any year in the decade, not the warmest. Every other year in the record averaged higher, ranging from 33.4C (2023) up to 34.5C (2024). The nine-year average excluding 2022 is 33.9C, meaning 2022 ran about 0.9C below the norm during its own monsoon season.

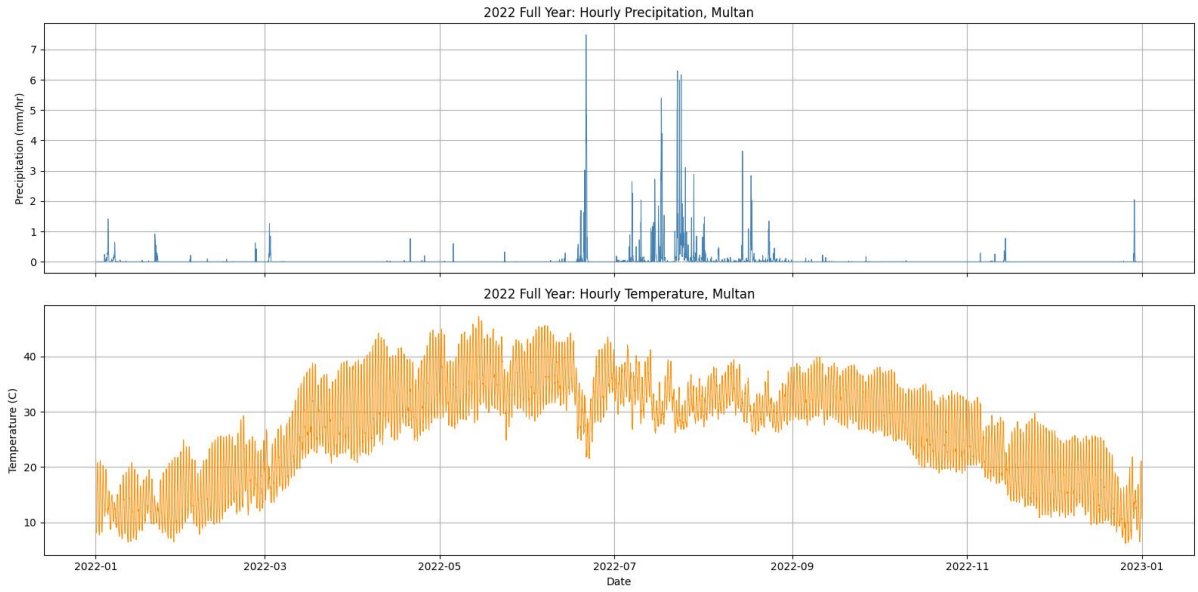


Figure 7. 2022 full-year hourly precipitation (top) and temperature (bottom), Multan.

Figure 7 resolves the apparent contradiction. Looking at the full year rather than only the monsoon-season average, 2022's highest temperatures - peaking above 44C - actually occurred in the pre-monsoon period (April through early June), before the rains arrived. Once the dense rainfall cluster begins in late June (visible in the top panel), the temperature swings in the bottom panel visibly narrow and shift downward - a clear, physically expected cooling effect from rain and cloud cover during the monsoon period itself.

Revised interpretation

2022 likely did feature an intense pre-monsoon heatwave, consistent with widely reported conditions across Pakistan that year - but this analysis, which averages temperature specifically across the monsoon months (June-September), captures the cooling effect of the heavy rain itself, not the pre-monsoon heat that may have helped set up the flood conditions. The two are not contradictory; they describe two different periods. A more precise test of the heat-precedes-flood mechanism would need to isolate April-May temperatures specifically and compare those across years, which is a natural extension of this analysis but was not part of its original scope.

5. Key Findings

- No statistically significant long-term trend in monsoon rainfall over 2016-2025 (p=0.591). The ten-year record is too short and too volatile to support a defensible trend claim - consistent with this project's own stated limitations.
- 2022 was an outlier in total rainfall volume and in the frequency of heavy-rain days, but not in single-day intensity - 2020 holds the record for the single wettest day in the dataset. This suggests 2022's flood risk came from sustained, repeated heavy rainfall (and the resulting ground saturation), rather than one extreme event.
- 2021 and 2022 bookend the decade as the driest and wettest years respectively. The dominant pattern in this record is volatility between extremes, not gradual, directional change.
- Rainfall peaks in the mid-to-late afternoon overall (15:00 PKT), but July and August show distinct peak-hour patterns (04:00 vs 15:00) that may reflect different underlying rainfall mechanisms.
- 2022's monsoon-season average temperature was the coolest of the decade, not the warmest - the flood period's heavy rain and cloud cover measurably cooled the season average, even though the pre-monsoon months of 2022 (outside this report's monsoon-only average) appear to have been unusually hot.

6. Limitations

- Short record: 10 years is short for formal climate trend detection (30+ years is the typical scientific standard); results describe recent variability, not a confirmed long-term climate signal.
- Spatial resolution: ERA5's native grid is approximately 25-31km - results represent a regional average around Multan, not a hyperlocal, station-level reading.
- No spatial/terrain analysis: flood routing, elevation, and drainage effects were deliberately outside this project's scope.
- No climate-model bias correction or CMIP6/CORDEX projection work was performed; this is an analysis of historical reanalysis data only, not a future projection.

7. Glossary of Key Concepts

Term	Plain-language meaning
Reanalysis	A model-plus-observations reconstruction of past weather, not a live measurement or forecast.
ERA5	The specific global reanalysis dataset (by ECMWF) used throughout this project.
CDS	Copernicus Climate Data Store - the free platform through which ERA5 was downloaded.
GRIB / NetCDF	Two different file formats for storing gridded climate data; this project used netCDF.
Accumulated variable	A value that builds up over time (e.g. rainfall); requires summing, not averaging.
Instantaneous variable	A value read at a single moment (e.g. temperature); requires averaging, not summing.

Term	Plain-language meaning
Bounding box / grid cell	A rectangular area defined by coordinates; ERA5 reports one value per ~25km grid cell within it.
Anomaly	How far a given year's value is from the multi-year average - positive means wetter/hotter than normal, negative means drier/cooler.
Linear regression / slope	A straight line fit through data over time; the slope is its rate of change per year.
R-squared	How much of the year-to-year variation the fitted trend line actually explains (0 = none, 1 = all).
P-value / statistical significance	The probability the observed trend could arise by chance alone; below 0.05 is conventionally called significant.
Percentile threshold	A cutoff defined by the data itself (e.g., the top 5% of all days), rather than a fixed number.
PKT / UTC	Pakistan Standard Time (UTC+5) vs. Coordinated Universal Time; all raw ERA5 timestamps are in UTC.

Appendix: Output Files

- daily_data.csv - full daily rainfall total and average temperature, all 1,473 days in the dataset.
- yearly_summary.csv - one row per year: total monsoon rainfall, wettest day, heavy-rain-day count, and longest dry spell.
- Multan_ERA5_Final_Analysis.ipynb - the complete, runnable Colab notebook underlying this report, including all code, charts, and printed statistics.

Both CSV files are saved in the project's Google Drive folder (ERA5_Multan) alongside the raw yearly netCDF files.